Wireless Networking

Task 1 : List 3 popular Wi-Fi standards (e.g., 802.11n, 802.11ac, 802.11ax) and for each, write one key difference in speed or frequency band.

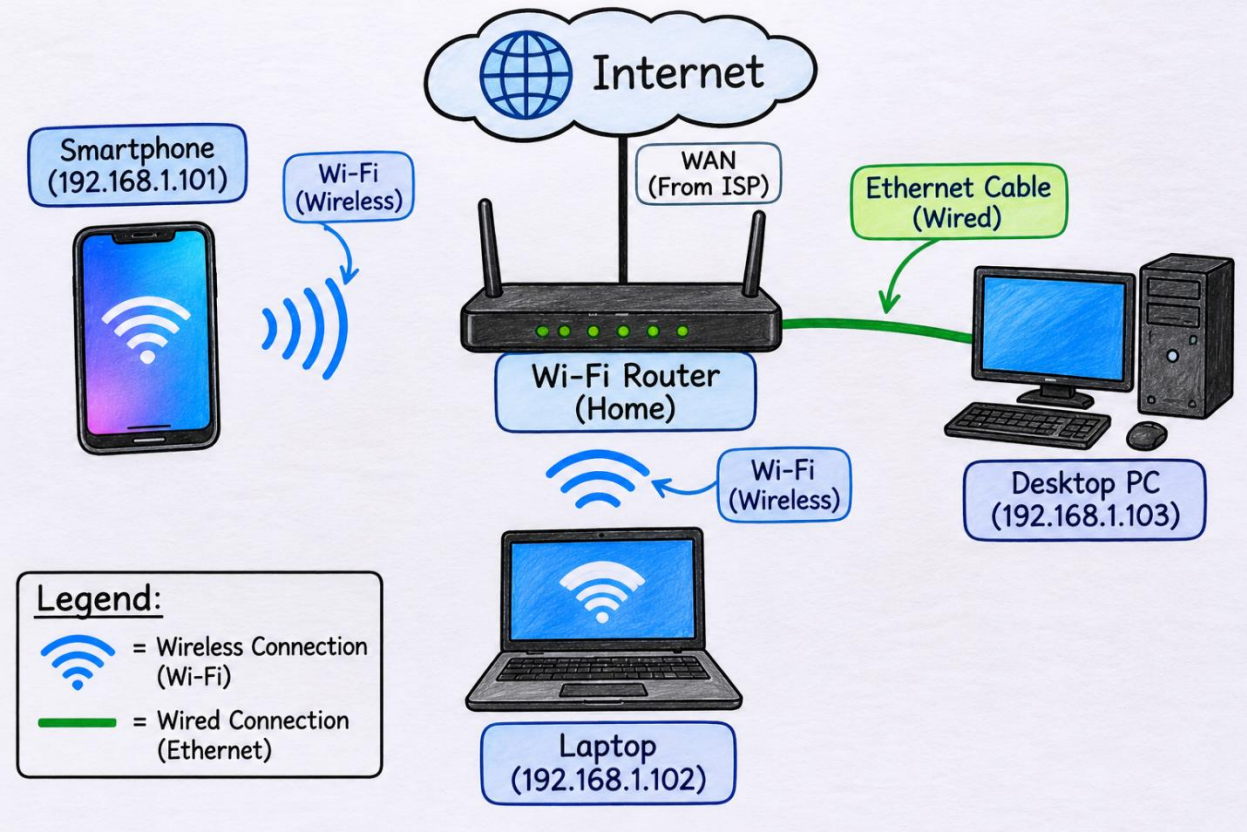
1. IEEE 802.11n (Wi-Fi 4):
Wi-Fi 4 operates on both 2.4 GHz and 5 GHz frequency bands and supports speeds up to 600 Mbps under ideal conditions. It is suitable for basic internet browsing, online classes, and video streaming.

2. IEEE 802.11ac (Wi-Fi 5):
Wi-Fi 5 mainly operates on the 5 GHz band and supports theoretical speeds up to approximately 6.9 Gbps. It offers faster connections and better performance for HD streaming, online gaming, and large downloads.

3. IEEE 802.11ax (Wi-Fi 6):
Wi-Fi 6 operates on both 2.4 GHz and 5 GHz bands and supports theoretical speeds up to approximately 9.6 Gbps. Its main advantage is improved efficiency when many devices are connected to the same network.

Conclusion:
Each new Wi-Fi standard improves speed and performance. Wi-Fi 4 is suitable for basic use, Wi-Fi 5 provides higher speeds, and Wi-Fi 6 handles multiple connected devices more efficiently.

Task 2 : Draw a simple diagram (on paper or using any drawing tool) showing how your smartphone connects to the internet via a Wi-Fi router at home, labeling the wireless and wired parts of the network.



A smartphone connects wirelessly to a home Wi-Fi router using Wi-Fi signals. The router receives the internet connection from the Internet Service Provider (ISP). The router forwards data between the smartphone and the internet.

The connection between the smartphone and the router is **wireless**, while the router's connection to the ISP may use a wired cable, such as Ethernet or fiber-optic cable.

Task 3 : Scan for available Wi-Fi networks using your laptop or phone, note down the SSID, signal strength, and security type (WPA2, WPA3, etc.) for at least 3 networks, and explain which one you would choose and why.

SSID (Network Name)	Signal Strength	Security Type
Kano 🎧 🖤 👀	85% (Strong)	WPA2-Personal
Sorath_PG1	85% (Strong)	WPA2-Personal
Airtel_Krish Heart care	48% (Medium)	WPA2-Personal

Network Selection: I would prefer a trusted Wi-Fi network with a strong signal and WPA2 or WPA3 security. A strong signal generally provides a more reliable connection, while security helps protect the network from unauthorized access. I would choose my own home network if it meets these requirements rather than connecting to an unknown network.

Conclusion: Scanning Wi-Fi networks helps us compare signal strength and security before choosing a network. A stronger signal and suitable security are important for a reliable and safer connection.

Task 4 : Explain in your own words what 'SSID hiding' is in Wi-Fi networks and discuss one advantage and one disadvantage of hiding your Wi-Fi SSID.

What is SSID Hiding?

- SSID stands for Service Set Identifier, which is the name used to identify a Wi-Fi network.
- SSID hiding is a feature in which a Wi-Fi router stops including its network name in normal Wi-Fi network broadcasts.
- As a result, the network name may not appear in the usual list of available Wi-Fi networks.
- To connect, users may need to enter the network name (SSID) manually along with the password.

Advantage of SSID Hiding:

- SSID hiding can make a Wi-Fi network less noticeable to casual users who scan for nearby networks.
- It may help reduce unwanted connection attempts from people who do not know the network name.

Disadvantage of SSID Hiding:

- SSID hiding does not make a Wi-Fi network completely invisible or secure.
- People using certain network-scanning tools may still discover it.
- It can also make connecting new devices more difficult because users must enter the network name manually.

Conclusion:

SSID hiding provides only a small amount of privacy and should not be considered a strong security measure. A Wi-Fi network should be protected with WPA2 or WPA3 encryption and a strong, unique password. These security measures are more important than hiding the network name.